MAITRI SHAH

Technical Portfolio Deep Dive | Research & Implementation in Trustworthy AI

I. CORE RESEARCH & WORK EXPERIENCE

1. Watermark Radioactivity Research (NeurIPS 2026 Submission)

Conducted at the **CISPA SprintML Lab**, this project investigates the persistence of data watermarks across multiple model generations—a phenomenon known as "radioactivity."

- **Technical Implementation:** Integrated high-resolution generative backends, including **SANA 0.6B** and **Infinity-2B**, into a unified pipeline for cross-architecture evaluation.
- **Manual Evaluation:** Performed foundational manual assessments of Infinity-2B to establish radioactivity baselines before the automated pipeline was operational.
- **Statistical Innovation:** Facilitated the transition from unstable P-values to **E-values**, providing non-asymptotic statistical guarantees for black-box provenance detection.
- **Knowledge Base:** Mastered the implementation and resilience testing of SOTA watermarking schemes such as **StegaStamp, TrustMark, RivaGAN, and BitMark**.

Key Learning: I learned how to manage distribution shifts in iterative training loops and why non-asymptotic metrics (E-values) are essential for reliable security claims in black-box scenarios.

2. Teaching Assistant | Trustworthy Machine Learning

Affiliated with **Saarland University and CISPA**, supporting a graduate-level cohort of 100+ students.

- **Co-leading Tutorials:** Designing and conducting technical walkthroughs on assignment logic and adversarial attack methodologies.
- **Infrastructure Automation:** Managing GPU cluster setups and troubleshooting SLURM-based job scheduling for large-scale student assignments.
- **Academic Development:** Co-developing course materials and evaluation criteria for exams and practical projects.

Key Learning: I developed the ability to translate complex research concepts into actionable engineering steps for others and learned to manage the technical infrastructure required for large-scale AI education.

3. QuadTransport (B.Tech Research Project)

- **Optimization:** Redesigned a Java-based implementation of Wasserstein distance computation using Quad-tree partitioning.
- **Impact:** Achieved a **50x runtime reduction**, reducing the execution window from 35 minutes to 12 seconds.

Key Learning: Deepened understanding of algorithmic efficiency and how spatial partitioning (Quad-trees) can drastically reduce the complexity of high-dimensional distance calculations.

II. EUROPEAN CHAMPIONSHIP IN TRUSTWORTHY AI

As a developer for this pan-European competition (200+ participants per venue), I built the technical infrastructure and challenge tasks that evaluated the next generation of AI security researchers.

Task 1: Image Attribution

Participants were required to classify images as originating from four specific models (VAR-16d, RAR-XL, Taming-cIN, Stable Diffusion) or as an outlier (OOD detection).

- **Metric:** Macro-averaged score weighted by ***TPR @ 1% FPR*** to enforce robustness against misclassifying natural images as generated.
- **My Contribution:** Developed the OOD detection logic and synthesized synthetic data generators for RAR and VAR models to provide participants with labeled training sets.

Key Learning: Understanding the "gap" between synthetic training distributions and real-world out-of-distribution detection, and how to calibrate models for extremely low false positive rates.

Task 2: Model Tracer

A white-box task identifying specific model architecture versions (e.g., VAR-d16 vs. VAR-d30) using model-level internal signals.

- **Challenge:** Differentiating between nearly identical model families by inspecting token probabilities, likelihoods, and loss values.
- **My Contribution:** Engineered the white-box access scripts (VAR/RAR generation) that exposed model-level signals beyond pixel-level outputs.

Key Learning: Learned that internal model activations and token-level likelihoods are far more reliable "fingerprints" for model attribution than the raw visual output alone.

Task 3: Watermark Detection (Vision)

Binary classification of clean vs. watermarked images across diverse embedding schemes.

- **Goal:** Designing a general-purpose detector capable of identifying signals regardless of the underlying embedding method.
- **Implementation:** Built a detection framework evaluated on continuous prediction scores [0, 1] at 1% FPR.

Key Learning: I mastered the statistical separation required to detect "invisible" noise and learned how different watermarking schemes (BitMark, Tree-Rings) leave distinct statistical markers.

Task 4: Watermark Removal (Vision)

An adversarial task focused on disrupting watermark signals while maintaining image utility.

- **Metric:** A combined score of watermark detectability (TPR) and image fidelity (MSE).
- **Implementation:** Designed the baseline attack scripts and developed the image quality mapping system that interpolated MSE values into an ImageQualityScore.

Key Learning: Explored the trade-off between privacy (signal removal) and utility (visual quality), learning how various data augmentations and noise patterns affect watermark resilience.

Task 5: LLM Watermark Detection

Identifying whether text samples were generated with watermarked token distributions.

- **Challenge:** Detecting signals across multiple LLM watermarking schemes (Kirchenbauer et al., Semantic Invariant, etc.).
- **Implementation:** Created the evaluation pipeline using ***TPR @ 1% FPR*** as the primary measure of detection robustness.

Key Learning: Learned the mechanics of green-list/red-list token biasing and how statistical z-scores are used to detect anomalies in text generation.

Task 6: LLM Watermark Removal

Modifying watermarked text to evade detection while preserving meaning and structure.

- **Metric:** A three-part product: detection evasion * token similarity (Levenshtein) * semantic similarity (Cosine).
- **Contribution:** Developed the combined evaluation score that prevented participants from simply paraphrasing the text entirely or submitting random strings.

Key Learning: Deepened understanding of semantic preservation and the mathematical difficulty of removing a signal embedded in token-level probability distributions.

Task 7: Infrastructure & Automation

- **Onboarding Scripts:** Developed Python/Shell automation for 200+ participants. The scripts handle dataset downloads from HuggingFace, environment configuration, and cluster path setups upon login.

Key Learning: Learned the critical importance of "Developer Experience" (DX) in technical competitions and how automation prevents large-scale setup failures.

III. SPECIALIZED RESEARCH: LLM UNLEARNING

Privacy-Preserving Unlearning on OLMo-2

A seminar project at CISPA focused on the removal of sensitive training data from large autoregressive models.

- **Methodology:** Evaluated Gradient Ascent (GA), GA with KL-Divergence (GA+KL), and Sequential GA.
- **Outcome:** Achieved a near-ideal **MIA AUROC of 0.529**, indicating high privacy protection.
- **Reality Check:** Analyzed the results to show that while forgetting was successful, it often came at the cost of model utility—a critical insight for future Grand Finale tasks.

Key Learning: Learned how to balance adversarial objectives with utility constraints and understood the fragility of fine-tuned privacy-preserving models.

IV. ACADEMIC CREDENTIALS

- **M.Sc. Data Science & AI | Saarland University:** CGPA: **1.8** (German Scale: 1.0 is highest, 4.0 is passing).
- **B.Tech. AI | NMIMS University (India):** CGPA: **3.95 / 4.0**.
- **Scholarship:** Deutschlandstipendium Scholarship Holder (2025–Present).